# Pascal Seitz

pascal_seitz@brown.edu | GitHub | LinkedIn

## EDUCATION

**Brown University** **Graduating May 2028**

Master of Science in Computer Science

- Courses: Machine Learning, Self-Supervised Learning and World Models

**Aalen University of Applied Sciences (Germany)** **Sep 2025**

Bachelor of Science in Information Systems, Concentrations: Applied ML and Software Engineering GPA: 3.8/4.0

- Courses: Data Mining and ML, Applied Deep Learning, Development of Information Systems, Software Engineering

## TECHNICAL SKILLS

- Programming languages: Python (TensorFlow, PyTorch, scikit-learn, Apache Spark, MLflow), Java, SQL, R
- Software: Git, Docker, Azure Databricks, Linux, Spring, Power BI

## PROFESSIONAL EXPERIENCE

**Data & Machine Learning Engineer Intern** **Mar 2026 – Aug 2026**

*Audi AG | Neckarsulm, Germany*

- Designed and implemented a production-ready end-to-end data pipeline in Databricks with Spark and Delta Lake, achieving a 17x runtime reduction and approximately 42% lower DBU costs
- Developed an LSTM-based anomaly detection model for automotive test-bench time-series data, with automated hyperparameter tuning to accelerate model experimentation

**Data Science Intern** **Sep 2025 – Mar 2026**

*TRUMPF | Ditzingen, Germany*

- Fine-tuned and extended a CNN-based computer vision model (“Cutting Assistant”) through data rebalancing and hyperparameter optimization, improving predictive accuracy while introducing a new cut-edge class
- Integrated a new multiclass cut-edge detection component, distinguishing edge presence, image orientation, alignment, and foreground/background conditions to improve robustness for downstream quality assessment

**Data Science Intern** **Sep 2023 – Feb 2024**

*Bosch Home Comfort Group | Wernau (Neckar), Germany*

- Led the development of an analysis platform, optimizing heat pump performance and enabling predictive reliability models, reducing annual costs by €800,000
- Collaborated with senior engineers to enhance predictive maintenance models using scikit-learn, improving system efficiency by 15% and reducing Scope 2 emissions

## RESEARCH EXPERIENCE

**Multimodal Dataset Infrastructure for ML Research | GalilAI Group, Brown University** **Oct 2026 – Present**

- Contributing new dataset integrations to stable-datasets, a multimodal research library designed for reproducible ML workflows across image, video, time-series, and audio data

**XAI for Anomaly Detection in Time-Series Data for Operating Diagnosis | Aalen UAS** **Jun 2025 – Sep 2025**

- Investigated explainable anomaly detection in multivariate industrial time-series data, developing an LSTM autoencoder to identify novel failure patterns and applying XAI methods to interpret model predictions
- Demonstrated early-warning potential for clogged-filter faults, identifying persistent anomaly patterns days to weeks before logged failures and translating them into sensor-level diagnostic insights using SHAP

**Heat Pump Field Data Analysis for Operating-State Detection and Load Profiling | Bosch** **Mar 2024 – Jul 2024**

- Applied the KDD framework and Python-based statistical analysis to investigate anomaly patterns in heat pump operational data, identifying a 30% system flow reduction caused by filter obstructions
- Co-authored a published paper on the research (DOI: 10.51202/9783181024379)

## LEADERSHIP & SERVICE

**Vice President of Education** **Jan 2022 – Jan 2023**

- Rebuilt the Education department of Germany’s largest student consulting network (JCNetwork; 2,600+ members) after the pandemic, leading a team of twelve fellows and coordinating educational initiatives across the network